



IILM UNIVERSITY

30-Days Data Analytics Internship

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Program & Semester:	Computer Science & Engineering (CSE)	Semester:	3rd Semester



Codtech IT Solutions Private Limited

About the Organization

Company Focus

Codtech IT Solutions Private Limited is dedicated to driving innovation and excellence in the realm of digital solutions. The company specializes in delivering top-notch web development and app development services tailored to meet unique client needs.

Core Mission

Empowering businesses through cutting-edge, scalable technology. Beyond core services, they are committed to nurturing the next generation of tech talent via comprehensive mentorship and structured internship programs.

Internship Objectives



Data Lifecycle

Understand the complete data lifecycle, from raw data extraction and processing to actionable business insights.



Tool Proficiency

Develop hands-on proficiency in Python, SQL databases, and interactive data visualization tools like Power BI and Tableau.



Predictive Modeling

Apply foundational machine learning techniques and statistical modeling directly to real-world business datasets.

Phase 1 - Exploratory Analysis

Foundation & Cleaning

- ✓ **Environment:** Utilized Python, Jupyter Notebook, Pandas, NumPy, and Seaborn.
- ✓ **Activities:** Cleaned raw datasets by effectively handling missing values and eliminating outliers using the IQR method.
- ✓ **Outcomes:** Generated comprehensive statistical summaries and correlation heatmaps to extract preliminary business insights.



Phase 2 - SQL Data Extraction

```

DATABASE Ecole;
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le;
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TABLE Etudiants (
INT AUTO INCREMENT PRIMARY KEY,
VARCHAR(50) NOT NULL,
nom VARCHAR(50) NOT NULL,
INT,
sse VARCHAR(10)
rows affected (0.02 sec)

r INTO Etudiants (Nom, Prenom, Age, Classe) VALUES ('Dupont', 'Marie', 21, 'A1'), ('Martin'
'B2'), ('Bernard', 'Sophie', 20, 'A1');
rows affected (0.01 sec)

E Etudiants SET Classe = 'B1' WHERE Prenom = 'Sophie' AND Nom = 'Bernard';
row affected (0.01 sec)

E FROM Etudiants WHERE Prenom = 'Paul' AND Nom = 'Martin';
row affected (0.01 sec)

T * FROM Etudiants;
+-----+-----+-----+
| Prenom | Age | Classe |
+-----+-----+-----+
| ont    | 21  | A1     |
| hard   | 20  | B1     |
+-----+-----+-----+
et (0.00 sec)

```

Relational Data Retrieval

- Environment: SQLite, PostgreSQL, and SQLAlchemy libraries.
- Activities: Integrated Python frameworks with robust SQL databases to script and automate data extraction protocols.
- Outcomes: Executed advanced analytical queries (CTEs, Window Functions) to uncover monthly sales trends and segment customers based on purchasing behavior.

Phase 3 - Visualizations

Dashboard Development

Leveraged Power BI and Tableau to transform raw datasets into intuitive, interactive executive dashboards. Integrated key performance indicator (KPI) cards to track critical metrics in real-time.

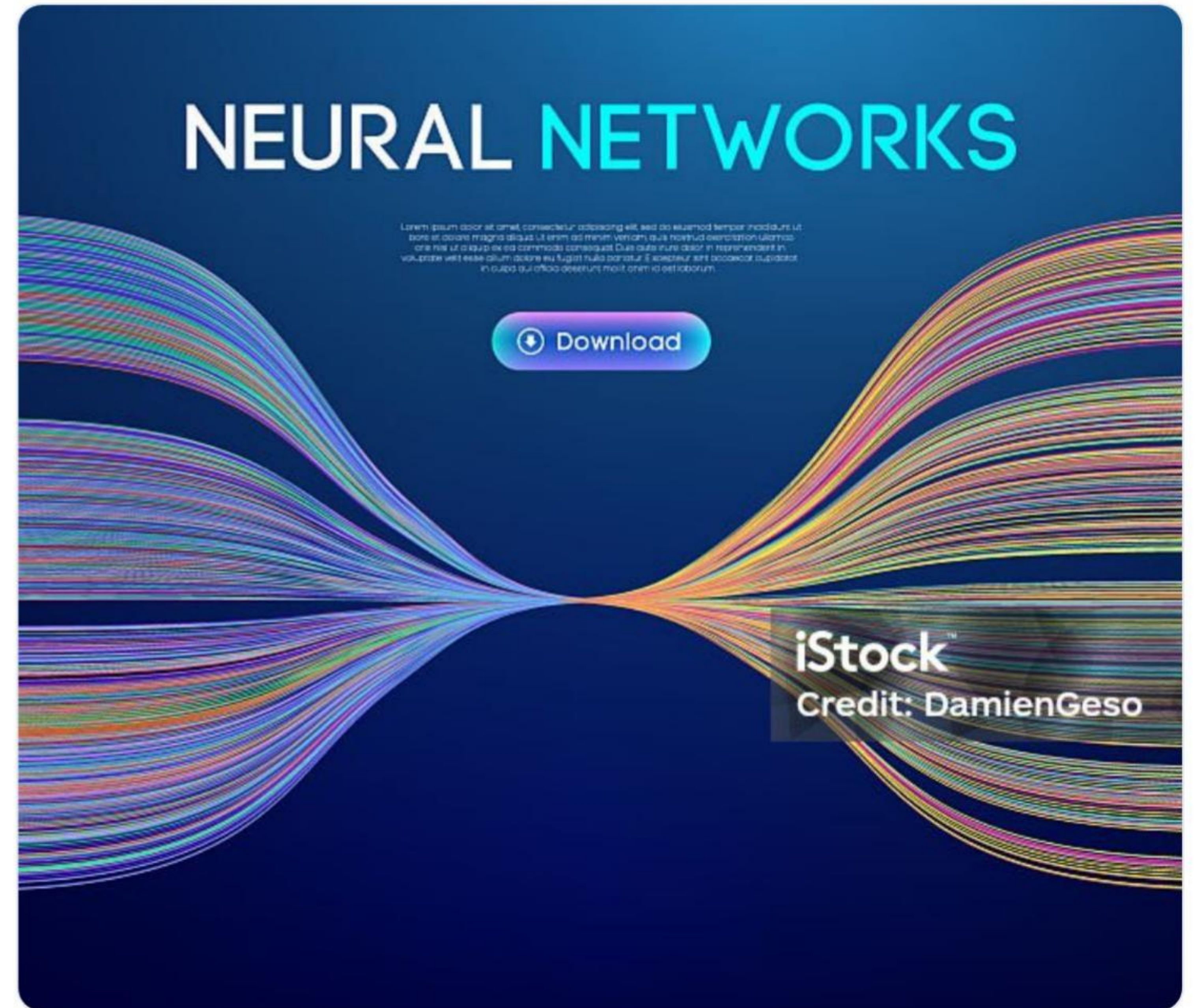
Visual Storytelling

Designed dynamic slicers, detailed category breakdowns, and geographic mapping to translate complex data tables into accessible, interactive visual stories optimized for stakeholder review.

Phase 4 - Advanced Analytics

Statistical & Predictive Modeling

- 📈 **Environment:** Employed Scikit-learn and SciPy for robust data modeling.
- 📈 **Activities:** Conducted strict hypothesis testing (t-tests, chi-square) and trained a K-Means clustering algorithm for targeted customer segmentation.
- 📈 **Outcomes:** Successfully deployed basic predictive models (Linear and Logistic Regression) to forecast critical target variables.



Phase 5 - Final Reporting



Automation & Executive Summary

-  Developed end-to-end Python scripts to autonomously load, clean, and process raw data on a defined schedule.
-  Compiled a comprehensive executive summary report highlighting top insights.
-  Provided actionable business recommendations based on evidence-backed data modeling.

Thank You

Avishek Yadav | IILM University

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